

1. What is the command to initialize a new Git repository?

```
PS E:\gitfirst> cd E:\
PS E:\> mkdir git-demo

Directory: E:\

Mode                LastWriteTime         Length
----                -
d-----          20-05-2026      23:06

PS E:\> cd git-demo
PS E:\git-demo> git init
```

2. What is the command to add files to the staging area?

```
PS E:\git-demo> echo "hello" > file.txt
PS E:\git-demo> git add file.txt
PS E:\git-demo> git add .
```

3. What is the command to commit changes with a message?

```
PS E:\git-demo> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   file.txt

PS E:\git-demo> echo "bye" > file2.txt
PS E:\git-demo> git add .
PS E:\git-demo> git commit -m "both files added"
[master (root-commit) 00b3fab] both files added
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 file.txt
 create mode 100644 file2.txt
PS E:\git-demo> git status
```

4. What is the command to check the status of the repository?

```
create mode 100644 file.txt
PS E:\git-demo> git status
On branch master
nothing to commit, working tree clean
```

5. What is the command to create a new branch and switch to it?

```
PS E:\git-demo> git checkout -b feature-branch
Switched to a new branch 'feature-branch'
PS E:\git-demo> git branch
* feature-branch
master
```

6. What is the command to merge a branch into the current branch?

```
PS E:\git-demo> echo "feature work" > feature.txt
PS E:\git-demo> git add .
PS E:\git-demo> git commit -m "added feature"
[feature-branch c6fc27d] added feature
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 feature.txt
PS E:\git-demo> git checkout main
error: pathspec 'main' did not match any file(s) known to git
PS E:\git-demo> git checkout
PS E:\git-demo> git merge feature-branch
Already up to date.
```

7. What is the command to view the differences between two branches or commits?

```
PS E:\git-demo> echo "updated content" >> file.txt
PS E:\git-demo> git add .
PS E:\git-demo> git commit -m "updated file"
[feature-branch 410b6a8] updated file
1 file changed, 0 insertions(+), 0 deletions(-)
PS E:\git-demo> git diff HEAD~1 HEAD
diff --git a/file.txt b/file.txt
index 2a93d00..68dd728 100644
Binary files a/file.txt and b/file.txt differ
```

8. What is the command to revert changes to a specific file?

```

PS E:\git-demo> git log --oneline
410b6a8 (HEAD -> feature-branch) updated file
c6fc27d added feature
00b3fab (master) both files added
PS E:\git-demo> echo "accidental change" >> file.txt
PS E:\git-demo> git restore file.txt
PS E:\git-demo> git restore --staged file.txt
PS E:\git-demo> echo "test" > file3.txt
PS E:\git-demo> git add .
PS E:\git-demo> git commit -m "test commit"
[feature-branch cfa0646] test commit
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 file3.txt

```

9. What is the command to undo the last commit?

```

PS E:\git-demo> git reset --soft HEAD~1
PS E:\git-demo> git reset --mixed HEAD~1
Unstaged changes after reset:
M   file.txt
PS E:\git-demo> git reset --hard HEAD~1
HEAD is now at 00b3fab both files added
PS E:\git-demo> git revert <commit-hash>
At line:1 char:12
+ git revert <commit-hash>
+ ~
The '<' operator is reserved for future use.
+ CategoryInfo          : ParserError: (:) [], ParentCo
ntainsErrorException
+ FullyQualifiedErrorId : RedirectionNotSupported

```

10. What is the command to push changes to a remote repository?

```
● PS E:\git-demo> git remote add origin https://github.com/PriyankaPatel1543/git-demo
● PS E:\git-demo> git remote -v
origin https://github.com/PriyankaPatel1543/git-demo (fetch)
origin https://github.com/PriyankaPatel1543/git-demo (push)
● PS E:\git-demo> git branch -M main
● PS E:\git-demo> git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 299 bytes | 149.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/PriyankaPatel1543/git-demo
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
● PS E:\git-demo> git push
Everything up-to-date
● PS E:\git-demo> git add .
● PS E:\git-demo> git commit -m "your message"
[main a5ac1c9] your message
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 file3.txt
● PS E:\git-demo> git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 330 bytes | 330.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/PriyankaPatel1543/git-demo
 00b3fab..a5ac1c9  main -> main
○ PS E:\git-demo> █
```